

Article

Comparative Hydrogen Production Routes via Steam Methane Reforming and Chemical Looping Reforming of Natural Gas as Feedstock

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Abstract: Hydrogen production is essential in the transition to sustainable energy. This study examines two hydrogen production routes, steam methane reforming (SMR) and chemical looping reforming (CLR), both using raw natural gas as feedstock. SMR, the most commonly used industrial process, involves reacting methane with steam to produce hydrogen, carbon monoxide, and carbon dioxide. In contrast, CLR uses a metal oxide as an oxygen carrier to facilitate hydrogen production without generating additional carbon dioxide. Simulations conducted using Aspen HYSYS analyzed each method's performance and energy consumption. The results show that SMR achieved 99.98% hydrogen purity, whereas CLR produced 99.97% purity. An energy analysis revealed that CLR requires 31% less energy than SMR, likely due to the absence of low- and high-temperature water-gas shift units. Overall, the findings suggest that CLR offers substantial advantages over SMR, including lower energy consumption and the production of cleaner hydrogen, free from carbon dioxide generated during the water-gas shift process.

Keywords: steam methane reforming; chemical looping reforming; hydrogen production; feedstock purification; natural gas



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1. Introduction

Hydrogen production can be achieved through various thermochemical processes, such as steam methane reforming (SMR), partial oxidation (POX), and auto thermal reforming (ATR), each involving the high-temperature conversion of methane into hydrogen, often with significant CO₂ emissions unless mitigated by carbon capture [1–4]. Chemical looping reforming (CLR) offers another efficient method, which uses a solid oxygen carrier to partially oxidize methane, producing syngas while managing heat effectively [5–7]. Several major companies, including Linde, Air Liquide, ThyssenKrupp, and others, produce hydrogen at an industrial scale using the SMR process [8]. For instance, Linde has constructed over 200 hydrogen plants with capacities ranging from 1000 to over 100,000 Nm³/h, whereas Air Liquide offers solutions for plants with capacities between 10,000 and 200,000 Nm³/h [9].

Hydrogen production began with natural gas pre-treatment to remove sulfur compounds, followed by steam generation. The core SMR process involves steam methane reforming at high temperatures using a heterogeneous catalyst to produce hydrogen (H₂) and carbon monoxide (CO), among other byproducts. The resulting reformer gas is then

subjected to a water–gas shift (WGS) reaction to convert CO into carbon dioxide (CO₂) and more H₂, followed by H₂ separation using pressure swing adsorption (PSA) to achieve high-purity hydrogen. Chemical-looping is a cyclic process that uses a solid metal oxide as an oxygen carrier, transferring oxygen from the air to the fuel through reduction–oxidation (redox) reactions, enabling the conversion of various feedstocks into valuable chemicals, such as ammonia, synthesis gas, and H₂/CO, at low production costs while addressing CO₂ capture. This process supplies the necessary heat for fuel conversion without the expense of oxygen production or mixing air with carbon-containing fuel gases. CLR partially oxidizes methane to produce high-quality syngas consisting of H₂ and CO. This process involves two reactors, where the oxygen carrier is alternately reduced in the fuel reactor and re-oxidized in the air reactor. The heat generated from the redox reactions is used to drive the endothermic reforming of methane [5].

Other hydrogen production from natural gas using thermochemical processes include ATR and POX, each with distinct advantages and challenges. ATR combines partial oxidation and steam reforming in a single reactor, allowing exothermic and endothermic reactions to balance [10]. This reduces the need for external heat, making the process more energy-efficient than standalone steam reforming. ATR is flexible, as it can operate with either air or pure oxygen, though using pure oxygen increases operational costs [11]. However, ATR still generates CO₂, necessitating additional steps for carbon management. POX directly reacts methane with a limited amount of oxygen, producing hydrogen and carbon monoxide without requiring steam [12]. This makes POX a faster, exothermic process that is less energy-intensive but produces lower hydrogen yields than ATR. POX is suited for small-scale or mobile applications due to its simpler setup and emits CO₂ [13]. Both ATR and POX offer advantages in specific contexts, with ATR being more energy-efficient for large-scale operations, whereas POX is valuable for quick, smaller hydrogen production setups. However, both methods still face environmental challenges due to CO₂ emissions, requiring integration with carbon capture technologies to reduce their carbon footprint.

The use of natural gas as feedstock for hydrogen production is absolute. Natural gas primarily consists of methane, CH₄ (90% by volume), light hydrocarbons (4–6% by volume), CO₂ (2% by volume), nitrogen (1.0% by volume), and hydrogen sulfide, H₂S (0.4% by volume). CO₂ reduces the density and calorific value of natural gas but is not toxic or corrosive like H₂S. H₂S is a product of decomposing sulfur-containing organic materials, and the biological reduction of sulfates poses several issues. It can corrode plant components, acidify lubricating oil in engines, and release sulfur oxides into the atmosphere, leading to acid rain [14]. Before utilization, natural gas must be purified to remove harmful and toxic compounds, including H₂S [15]. Different feedstock purification technologies come with various advantages and drawbacks. Although efficient in CH₄ recovery, water absorption involves high costs, energy consumption, and water requirements, with issues such as bacterial growth and foaming [16,17]. Organic solvent absorption is efficient but costly and complex, requiring expensive solvents and high energy for regeneration [18]. Chemical absorption ensures high CH₄ recovery and complete H₂S removal but entails high investment, energy needs, and potential for chemical fouling and corrosion [19]. PSA offers high CH₄ recovery and is suitable for small capacities but requires prior elimination of H₂S and moisture [20]. Cryogenic separation provides high-purity CH₄ and CO₂ but involves expensive, energy-intensive processes [21]. Although cost-effective and straightforward, membrane separation often necessitates multiple stages for high purity and has significant membrane maintenance needs [22]. The efficiency and applicability of each method heavily depend on specific operational and economic conditions.

With high-purity hydrogen being produced via SMR and CLR, these methods play a crucial role in supplying hydrogen for energy generation and transportation applications. One of its most prominent uses is as a clean fuel, particularly for fuel cell vehicles (FCVs), where hydrogen combines with oxygen in a fuel cell to generate electricity, with water vapor as the only emission [23,24]. Additionally, hydrogen is vital in industrial processes, including ammonia production for fertilizers, refining petroleum, and synthesizing

methanol [25,26]. Hydrogen can also be utilized in power generation, either directly in hydrogen turbines or through blending with natural gas [27–29]. As global interest in green hydrogen increases, hydrogen has significant potential to act as an energy carrier in sectors such as heavy transportation, aviation, and even long-term energy storage [30]. This wide range of applications underscores hydrogen's importance in reducing carbon emissions and advancing sustainable energy solutions.

This work proposes the production of cleaner hydrogen using SMR and CLR utilizing purified natural gas. A critical component in this process is selecting an appropriate feed-stock purification system to ensure the efficiency and effectiveness of hydrogen production. Performance analysis was performed using Aspen software to determine the effectiveness of the purification system chosen for SMR and CLR.

2. Natural Gas Purification System

The solubility of different gas fractions in a liquid-absorbing solution is the basis of the physical absorption principle. The components of the absorbed gas form a physical link with the absorbing liquid [31]. During physical absorption, H_2S is removed in water or organic solvents. In contrast, for the chemical absorption process, the water solubility of H_2S is enhanced using alkaline water or by oxidizing H_2S into more water-soluble compounds [32]. Water is the most widely used and most straightforward solution. However, this method requires high water consumption and regeneration or a huge amount of fresh water to avoid regeneration [33]. An alternative to water is the use of organic solvents. Smaller volume consumption compared to water makes the size of the facility more compact [17]. The most common organic absorbent is Selexol (made from dimethyl ether of polyethylene glycol), which exhibits an affinity for H_2S several times greater than water [34]. The rectisol process uses methanol to improve absorption; however, due to the necessary cooling (approximately $-40\text{ }^{\circ}\text{C}$), this process is costly and requires more energy for regeneration than water [17,34].

In the chemical absorption process, the absorption of H_2S in solutions containing alkalis (such as sodium hydroxide ($NaOH$) and potassium hydroxide (KOH)), iron (Fe) compounds (such as iron(II) chloride ($FeCl_2$) and iron(III) oxide-hydroxide ($Fe(OH)_3$)), strong oxidants (such as hydrogen peroxide (H_2O_2) and magnesium peroxide (MgO_2)), and alkanolamines. The most popular alkanolamines include monoethanolamine (MEA), diethanolamine (DEA), and N-methyldiethanolamine (MDEA). The advantages of alkanolamines include their relatively low cost, good thermal regenerability, and low affinity for CH_4 [34]. Fe-compound absorbents have been commercially applied in France, Poland, and Florida for large plants, but they have been reported to be economically unsustainable [35].

Adsorption occurs on the surface. The selective natural gas contaminants' adsorption process uses various porous adsorbent materials [36]. Typical adsorbents include activated carbon, iron oxide, iron hydroxide, or zinc oxides, such as Sulphur-Rite[®] (GTP-Merichem, Schaumburg, IL, USA), SulfaTreat[®] (MILLC, Fenton Township, MI, USA), SOXSIA[®] (Gastreatment. Services B.V., GV Bergambacht, The Netherlands), Media-G₂[®], and Sulfa-Bind[®] (ADI International Inc., Brookfield, PE, Canada). The basic principle is that H_2S reacts readily with iron or zinc oxides to form insoluble iron sulfide (FeS) or zinc sulfide (ZnS), respectively [14]. Pressure swing adsorption (PSA) is widely used in adsorption [37]. Generally, the PSA separation of gas mixtures consists of four steps: adsorption (selective adsorption of CO_2 and/or O_2 and/or N_2 , producing pure CH_4), decompression/depressurization (depressurization of the adsorption vessel to the atmospheric pressure), desorption/regeneration (the purified CH_4 is recycled to displace CO_2 to reduce CO_2 content, but this results in CH_4 loss), and pressurization [36,38]. Higher pressure corresponds to higher gas adsorption. Impurities in raw natural gas may influence the effectiveness of this method; therefore, supplementary H_2S removal is needed beforehand [39].

Cryogenic separation technology uses the principle of liquefying different gases under high pressures and low temperatures [36]. To prevent icing and equipment blockages, H_2S

and water in the feedstock must be treated beforehand, making this method unsuitable for H₂S removal. Although it can yield highly pure natural gas, it demands extensive instrumentation, including heat exchangers, distillation columns, turbines, and compressors. This results in high capital and operating expenses since it requires a large amount of energy to cool and solidify CO₂ at low temperatures [30]. Using a thin-film porous barrier, usually made of interconnected hollow plastic fibers, the membrane separation method selectively diffuses natural gas through the barrier. This process occurs due to differences in concentration and/or pressure on either side of the membrane and is most suitable for high H₂S concentration applications [34]. Gases have varying permeabilities; therefore, the membrane fiber permits partial permeability of oxygen and H₂S and only minimal passage of N₂ and CH₄ [40]. Table 1 compiles recent commercial purification systems for natural gas.

Table 1. Commercial purification systems used for natural gas [15,16,34,41–53].

System	H ₂ S Removal	CO ₂ Removal	CH ₄ Purity, %	CH ₄ Loss, %	TRL	Commercial
Physical absorption						
Water absorption	•	•	Up to 98	1–4	9	• Condorchem Envitech (Barcelona, Spain), Durr (Stuttgart, Germany)
Organic absorption	•	•	Up to 98	1.5–4	9	• Selexol™ (Dow, Midland, MI, USA), Rectisol® (Linde, Dublin, Ireland), BgPur™ (Eco-Tec Inc., Pickering, ON, Canada)
Chemical absorption						
NaOH aqueous solution	•	•	>94	<0.1	9	• Sulfurex® CR (DMT Clear Gas Solutions, Tualatin, OR, USA)
Iron (Fe)-chelated/ethylene-diamine-tetraacetic acid (EDTA) solutions	•	•	Up to 98	1–3	9	• Biosulfex (Innovative Group PROMIS, Warszawa, Poland), SulFerox® (Shell/Dow), Lo-Cat® (Houston, TX, USA Filter/Merichem), Sulfothane® (Veolia Water Technologies, Pennsauken, NJ, USA)
Alkanolamines	•	•	>99	0.04–0.5	9	• Mercedes Ventures LLC Inc. (New York, NY, USA), HZI Biomethan (Zurich, Switzerland)
Biological absorption	•	•	Up to 98	<10	8–9	• THIOPAQ® (Shell-Paques, AD Utrecht, The Netherlands)
Membrane Separation	•	•	Up to 99	0.1–5	9	• Durr (Stuttgart, Germany), HZI Biomethan (Zurich, Switzerland)
Cryogenic Separation		•	>97	<2	9	• Scandinavian GtS (Copenhagen, Denmark), FirmGreen (Newport Beach, CA, USA)
Pressure swing adsorption (PSA)	•	•	Up to 99	1–4	9	• Schmack-Biogas AGCarboTech (Schwandorf, Germany), Guild Associates, Inc. (Dublin, OH, USA)

Amine absorption is a widely preferred method for purification due to its high selectivity and efficiency in removing acidic gases, such as CO₂ and H₂S. Amines exhibit a strong affinity for these acidic components, ensuring effective and selective removal, resulting in high-purity CH₄. The scalability of amine absorption systems makes them adaptable for various operational scales, accommodating both small and large purification plants. Furthermore, amines can be regenerated and reused multiple times, enhancing the economic and sustainable aspects of the process [54,55]. The absorbed CO₂ and H₂S are typically desorbed by heating the amine solution, which can then be cycled back into the absorption process [56–58]. This regeneration capability reduces operational costs and minimizes the environmental impact, making amine absorption a practical and efficient choice among commercial purification systems [59–61]. Therefore, amine absorption is selected to purify natural gas for hydrogen production using SMR and CLR.

3. System Modelling

The SMR process creates hydrogen and carbon dioxide by reacting natural gas and steam at high temperatures [62]. This method is well-established and widely utilized due to its high efficiency and scalability. CLR offers a promising alternative by integrating a metal oxide-based oxygen carrier, facilitating the indirect oxygen transfer from air to the hydrocarbon feedstock [9]. These systems utilize natural gas, contributing to a more sustainable and renewable hydrogen production pathway. To analyze and compare these hydrogen production systems, Aspen HYSYS v14.0 is employed for process modeling and simulation. The software enables detailed design and optimization of SMR and CLR processes to achieve a hydrogen production rate of 500 kg/h. Simulating the entire process flow, from feedstock input to hydrogen output, helps to identify critical operational parameters and potential bottlenecks. This modeling approach can thoroughly evaluate the comparative performance and energy consumption of hydrogen production from raw natural gas via SMR and CLR.

3.1. Steam Methane Reforming

Steam methane reforming process:

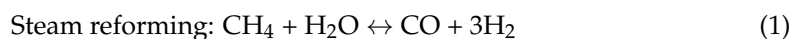


Figure 1 shows the process flow for the SMR process using natural gas as the feedstock, based on the established hydrogen plant by Linde [63], including the following processes:

1. Amine absorption for natural gas purification: The compressor has an inlet pressure of 2000 kPa and an outlet pressure of 3000 kPa [64,65]. The inlet and outlet temperatures of the heater are 57.99 °C and 101 °C, respectively [65]. The absorber has a top pressure of 500 kPa and a bottom pressure of 3000 kPa, with 40 stages [66]. The mixer has an inlet and outlet temperature of 66.97 °C and 75 °C, respectively [65]. The lean amine and make-up water have similar flow rates of 12.50 kgmol/h. The cooler reduces the acid gases to room temperature for amine regeneration.
2. Steam methane reforming: The natural gas and the demineralized water were heated to 1000 °C [63]. The reformer had a pressure of 500 kPa, with a reaction frequency factor of 2.0×10^{-3} and an activation energy of 109.40 kJ/mol [63,65–67].
3. Water gas shift for CO conversion: The reformer gases and the water vapor were heated to 450 °C. The shifter had a pressure of 500 kPa, with a reaction frequency factor of 2.8×10^{-2} and an activation energy of 100 kJ/mol [63,65–67].
4. PSA for hydrogen purification: the shifted reformer gases were cooled to 25 °C and then purified at 500 kPa [63,65].

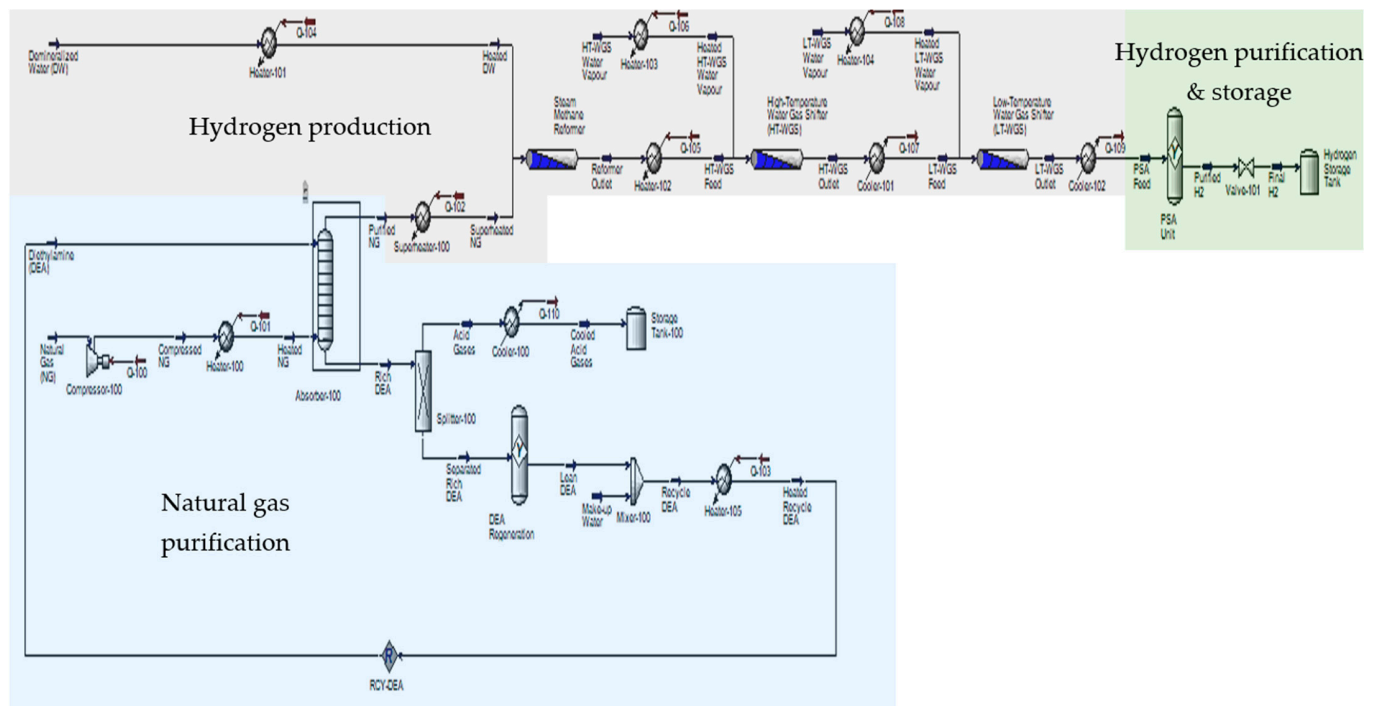


Figure 1. The steam methane reforming process with a natural gas purification system.

3.2. Chemical Looping Reforming

Chemical looping reforming process:

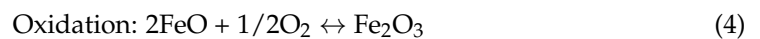
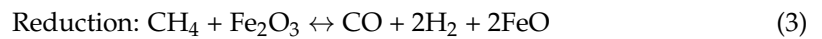


Figure 2 shows the process flow for the CLR process using natural gas as the feedstock, including the following processes:

1. Amine absorption for natural gas purification: The compressor has an inlet pressure of 2000 kPa and an outlet pressure of 3000 kPa [63,64]. The compressed natural gas is heated to 101 °C [64]. The absorber has a top pressure of 500 kPa and a bottom pressure of 3000 kPa, with 40 stages [65]. The lean amine and make-up water have similar flow rates of 12.5 kgmol/h. The cooler reduced the acid gases to room temperature for amine regeneration.
2. Chemical looping reforming: The pressurized natural gas was heated to 1000 °C [68]. The fuel reformer reduced CH_4 with 99% conversion at 800 °C and 500 kPa for hydrogen production [65]. The oxygen carrier was oxidized with 100% conversion at 720 °C and 500 kPa for regeneration [69].
3. PSA for hydrogen purification: the shifted reformer gases were cooled to 25 °C and were then purified at 500 kPa [63,65].

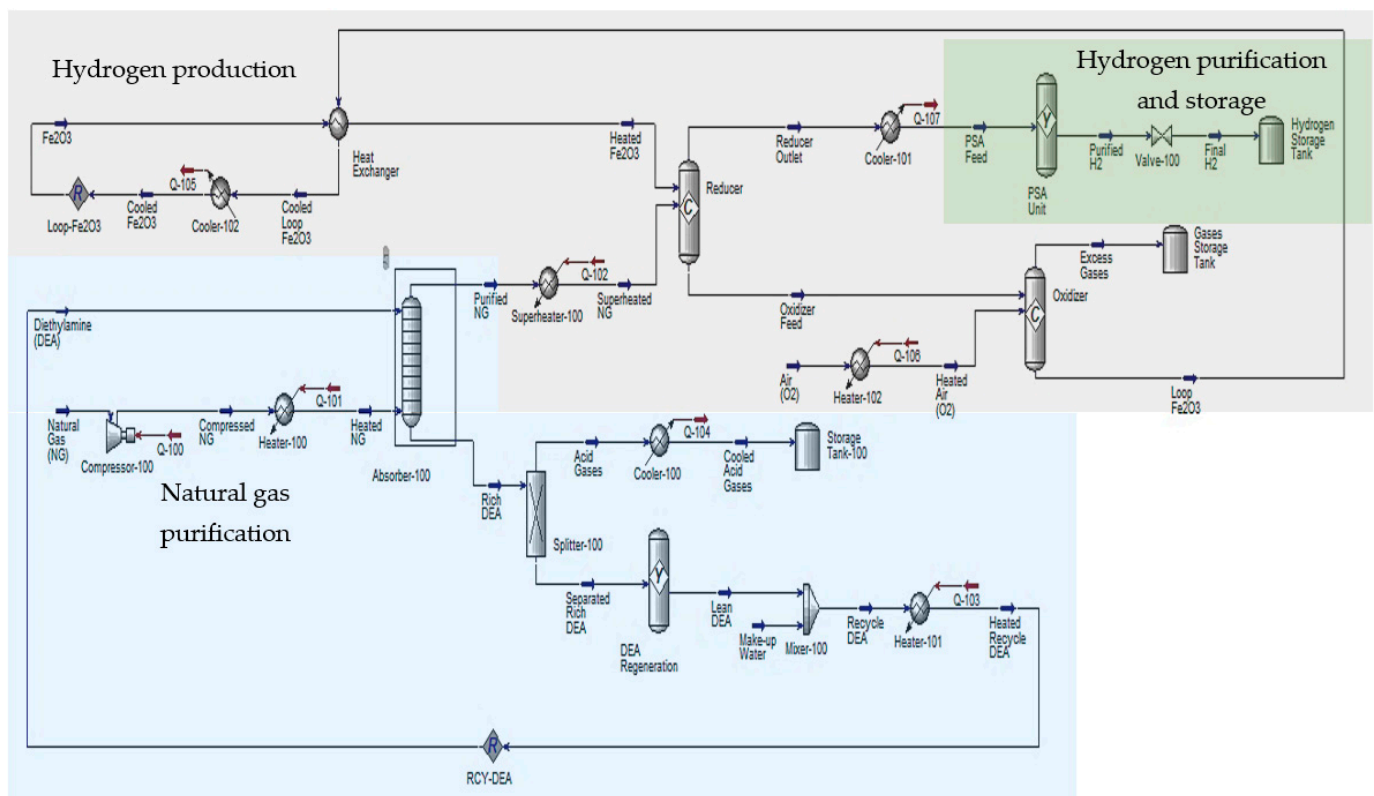


Figure 2. Chemical looping reforming with a natural gas purification system.

4. Performance and Energy Analysis

4.1. Comparison of Hydrogen Production by Different Processes

Raw natural gas mainly comprises of methane, hydrogen sulfide, small amounts of nitrogen, and light hydrocarbons, such as ethane, propane, and butane. To prepare this natural gas for SMR and CLR, it must first be purified to remove contaminants that can impede these reactions. An essential step in this purification process is using an amine absorption system, eliminating acidic gases such as hydrogen sulfide and carbon dioxide from the raw natural gas. The amine absorption system passes the natural gas through a solution containing amines. These amines chemically react with H_2S and CO_2 , forming compounds that can be separated from the gas stream. This purification step ensures that the downstream processes run more efficiently and produce higher hydrogen yields with fewer contaminants.

The endothermic reaction of SMR produces hydrogen and carbon monoxide with residual methane, water, and small amounts of ethane, butane, propane, and nitrogen. This reaction is represented by Equation (1). Carbon monoxide combines with more steam in a water–gas shift process to produce more hydrogen and carbon dioxide, as described by Equation (2). Nevertheless, this procedure uses a large amount of energy and produces carbon dioxide, which is removed during feedstock purification [70,71]. Referring to Table 2, methane, carbon dioxide, nitrogen, traces of light hydrocarbons, and water are still contained in the shifted reformer gases. PSA is used to purify the hydrogen produced [72]. Basin et al. [73] further explained that the compressor and dryer of PSA (step 1) removed water and methane, the cold box of PSA (step 2) separated carbon dioxide, and the membrane of PSA (step 3) separated the trace gases. Therefore, cleaner hydrogen is produced, since the end product has no carbon dioxide content.

Table 2. Molar compositions at the inlet of each unit of SMR and CLR.

Unit	Compound		Molar Composition at the Inlet	
			SMR	CLR
Raw natural gas (Department of Standards Malaysia, 2011a) ¹	Methane	CH ₄	0.9046	0.9046
	Ethane	C ₂ H ₆	0.0419	0.0419
	Propane	C ₃ H ₈	0.0101	0.0101
	Butane	C ₄ H ₁₀	0.0066	0.0066
	Carbon Dioxide	CO ₂	0.0226	0.0226
	Hydrogen Sulfide	H ₂ S	0.0042	0.0042
	Nitrogen	N ₂	0.0100	0.0100
	Water	H ₂ O	0.0000	0.0000
	Hydrogen	H ₂	0.0000	0.0000
	Carbon Monoxide	CO	0.0000	0.0000
Amine absorption (purified natural gas)	Methane	CH ₄	0.8985	0.9005
	Ethane	C ₂ H ₆	0.0415	0.0416
	Propane	C ₃ H ₈	0.0100	0.0100
	Butane	C ₄ H ₁₀	0.0065	0.0066
	Carbon Dioxide	CO ₂	0.0000	0.0002
	Hydrogen Sulfide	H ₂ S	0.0000	0.0000
	Nitrogen	N ₂	0.0042	0.0042
	Water	H ₂ O	0.0393	0.0370
	Hydrogen	H ₂	0.0000	0.0000
	Carbon Monoxide	CO	0.0000	0.0000
Hydrogen production reforming	Methane	CH ₄	0.1955	0.0065
	Ethane	C ₂ H ₆	0.0152	0.0150
	Propane	C ₃ H ₈	0.0037	0.0036
	Butane	C ₄ H ₁₀	0.0024	0.0024
	Carbon Dioxide	CO ₂	0.0000	0.0001
	Hydrogen Sulfide	H ₂ S	0.0000	0.0000
	Nitrogen	N ₂	0.0015	0.0015
	Water	H ₂ O	0.2451	0.0134
	Hydrogen	H ₂	0.4024	0.6383
	Carbon Monoxide	CO	0.1341	0.3192
HT-WGS	Methane	CH ₄	0.1946	-
	Ethane	C ₂ H ₆	0.0152	-
	Propane	C ₃ H ₈	0.0037	-
	Butane	C ₄ H ₁₀	0.0024	-
	Carbon Dioxide	CO ₂	0.1255	-
	Hydrogen Sulfide	H ₂ S	0.0000	-
	Nitrogen	N ₂	0.0015	-
	Water	H ₂ O	0.1231	-
	Hydrogen	H ₂	0.5261	-
	Carbon Monoxide	CO	0.0080	-
LT-WGS	Methane	CH ₄	0.1938	-
	Ethane	C ₂ H ₆	0.0151	-
	Propane	C ₃ H ₈	0.0036	-
	Butane	C ₄ H ₁₀	0.0024	-
	Carbon Dioxide	CO ₂	0.1327	-
	Hydrogen Sulfide	H ₂ S	0.0000	-
	Nitrogen	N ₂	0.0015	-
	Water	H ₂ O	0.1193	-
	Hydrogen	H ₂	0.5314	-
	Carbon Monoxide	CO	0.0003	-

Table 2. Cont.

Unit	Compound		Molar Composition at the Inlet	
			SMR	CLR
PSA (end product)	Methane	CH ₄	0.0000	0.0000
	Ethane	C ₂ H ₆	0.0000	0.0000
	Propane	C ₃ H ₈	0.0000	0.0000
	Butane	C ₄ H ₁₀	0.0000	0.0000
	Carbon Dioxide	CO ₂	0.0000	0.0000
	Hydrogen Sulfide	H ₂ S	0.0000	0.0000
	Nitrogen	N ₂	0.0002	0.0003
	Water	H ₂ O	0.0000	0.0000
	Hydrogen	H ₂	0.9998	0.9997
	Carbon Monoxide	CO	0.0000	0.0000

¹ This Malaysian Standard was developed by the Technical Committee on Natural Gas under the authority of the Petroleum and Gas Industry Standards Committee.

In CLR, the process produced syngas with higher hydrogen purity, lower methane, light hydrocarbons, and a significant amount of carbon monoxide from the reaction. Pouw et al. [74] suggested that hydrogen production using CLR can be performed without WGS. This is because the endothermic reactions of SMR and WGS are similar to those in CLR's fuel reactor. The results in Table 2 show that the hydrogen yield produced from the CLR process was higher than that produced by the SMR process, which emphasized the need for WGS in the SMR process compared to the CLR process. Dumbrava and Cormos [75] reported that the CLR process does not require further WGS because high-purity hydrogen can be produced. CLR is an alternative method for hydrogen production and has also been proposed as one of the carbon capture systems [76]. This has been proven from the results in Table 2, where the CLR process using natural gases produced high-purity hydrogen with trace gases, including unconverted carbon monoxide. There is no acid gas content in the produced syngas. The resulting gas mixture was then passed through a series of purification steps to isolate high-purity hydrogen, with a small amount of nitrogen as a byproduct. Dumbrava and Cormos [75] reported that PSA is unnecessary, but many studies, such as Cao et al. [77], report that PSA is needed in the hydrogen production process to purify hydrogen. The PSA unit can adsorb all unconverted methane and carbon monoxide to increase hydrogen yield [78]. From these, the produced hydrogen can be categorized as cleaner hydrogen.

4.2. Analysis of Energy Consumption

When considering industrial-scale hydrogen production plants, the energy consumption figures directly translate into electricity usage. The required energy for each energy stream is shown in Table 3. The inlet flow rate for raw natural gas is 800 kgmol/h. In the reforming process, a significant amount of energy is required to heat the reactants to the necessary temperatures for efficient chemical reactions [79]. In particular, SMR requires heating water to temperatures as high as 1000 °C from room temperature to initiate the reaction. This process requires substantial energy input because water has a high specific heat capacity, meaning it takes considerable energy to raise its temperature. In contrast, the CLR process involves heating air in the air reactor from around 15 °C to temperatures typically up to 800 °C. The specific heat capacity of air is lower than that of water, and the required temperature increase in CLR is generally less. Therefore, the energy required to heat air is significantly lower than that needed to heat water in SMR. The heat capacity of water is much higher than that of air [80]. Heating water requires raising its temperature and converting it to steam, which consumes substantial energy [81]. In contrast, heating air involves only increasing temperature without a phase change, making it less energy intensive. This explains the lower energy requirements in the chemical looping reforming process.

Table 3. Generated required energy for each stream.

Unit	Component	Required Energy (kWh)	
		SMR	CLR
Amine absorption	Compressor	Q-100 291.11	Q-100 291.11
	Feedstock heater/cooler	Q-101 406.39	Q-101 406.39
	Recycle amine heater	Q-103 184.58	Q-103 150.33
	Acid gases cooler	Q-110 3.53	Q-104 3.99
Steam methane/chemical looping reforming process	Superheater	Q-102 13,336.11	Q-102 9741.67
	Demineralized water/air heater	Q-104 18,247.22	Q-106 6330.56
	Oxygen carrier cooler	-	Q-105 2198.61
High-temperature water–gas shift	Reformer gases heater	Q-105 3894.44	-
	Water vapor heater	Q-106 164.94	-
	Reformer gases cooler	Q-107 7691.67	-
Low-temperature water–gas shift	Water vapor heater	Q-108 144.25	-
Pressure swing adsorption	Shifted reformer gases/reformer gases cooler	Q-109 8083.33	Q-107 16,919.44
Total		52,447.59	36,042.11

It can also be observed that CLR requires an additional energy stream to manage the heat from the oxygen carrier cooler. This additional cooling step is necessary due to the exothermic reactions involving the oxygen carriers, which release significant heat that must be removed to maintain optimal reaction conditions [82]. Despite this added complexity, CLR's lack of a WGS reaction to enhance hydrogen yield presents a notable advantage. The absence of WGS in CLR simplifies the process and reduces the overall energy demand associated with shifting the reaction equilibrium towards higher hydrogen production. Furthermore, during hydrogen purification, the higher energy requirement for PSA in CLR may be attributed to the higher hydrogen-to-carbon-monoxide (H_2/CO) ratio compared to the SMR–WGS process. In CLR, the direct oxidation of the fuel results in a gas mixture with a higher H_2/CO ratio, which necessitates more energy-intensive separation processes in PSA to achieve the desired hydrogen purity levels [83]. The PSA process in CLR must handle this richer hydrogen stream, which can be more demanding in terms of energy and adsorbent material cycling. This leads to increased energy consumption compared to SMR.

Figure 3 shows that the lowest total energy requirement was for CLR. This is primarily due to the elimination of the WGS unit, which is necessary for steam methane reforming processes. The CLR process simplifies the hydrogen production pathway by directly utilizing natural gas and oxygen carriers, reducing the need for additional energy-intensive components and steps. The advantages of the CLR process in terms of energy consumption are significant. The absence of acid gas formation, such as CO_2 , during the hydrogen production process further highlights the environmental benefits of CLR. From the overall results, it can be concluded that the lower energy demands and cleaner output of the CLR

process make it a superior choice for green hydrogen production, validating its efficiency and environmental benefits over traditional SMR.



Figure 3. Total energy consumption for all processes.

SMR frequently requires extensive post-reaction purification processes, such as the WGS reaction, which increases both energy consumption and process complexity while generating carbon dioxide, which must be managed carefully. In contrast, the implementation of an amine absorption system for the pre-purification of natural gas, specifically focusing on feedstock (natural gas) purification, plays a vital role in enhancing overall efficiency. Research conducted by Dumbrava and Cormos [75] indicates that high-purity hydrogen can be produced without additional purification stages, emphasizing the significance of pre-purification in improving process effectiveness. Although previous studies often highlight the necessity of WGS to achieve cleaner hydrogen, current findings underscore CLR's ability to utilize the inherent properties of the reforming reaction to produce syngas with minimal contaminants. This points to the critical importance of an effective natural gas purification process, which significantly influences the performance and sustainability of hydrogen production. Although there may be an increase in energy consumption during the PSA process of CLR, due to the higher H_2/CO ratio in the gas mixture, the overall energy requirements of CLR remain lower than those of SMR. This positions CLR as a more streamlined and energy-efficient alternative for hydrogen production, yielding cleaner hydrogen with fewer byproducts. Consequently, effective natural gas purification has emerged as a key element in addressing the limitations of other hydrogen production methods, such as SMR, ATR, and POX.

5. Conclusions and Outlook

Raw natural gas primarily contains methane, hydrogen sulfide, and small amounts of nitrogen and light hydrocarbons, such as ethane, propane, and butane. Purification of the raw natural gas is necessary to remove contaminants that hinder superfluous reactions in SMR and CLR. An essential purification step involves using an amine absorption system to eliminate acidic gases, such as hydrogen sulfide and carbon dioxide. The endothermic SMR process produces hydrogen and carbon monoxide, with residual methane, water, and minor amounts of ethane, propane, butane, and nitrogen, which are further processed through a water–gas shift reaction to increase hydrogen yield but at a high energy cost. This energy-intensive process, combined with the need for additional purification, makes SMR less efficient than CLR. CLR simplifies hydrogen production by eliminating the WGS reaction, which reduces energy consumption and avoids acid gas formation, leading to cleaner hydrogen output. Additionally, the CLR process requires less energy to heat air than

water in SMR, further emphasizing its efficiency. The findings indicate that when using natural gas, SMR produced hydrogen with 99.98% purity and CLR produced hydrogen with 99.97% purity, with 0.002% and 0.003% nitrogen as byproducts, respectively. The energy consumption analysis shows that the CLR system requires 31% less energy than SMR, possibly due to the absence of the low-temperature and high-temperature WGS units, making CLR an excellent alternative route for producing cleaner hydrogen from natural gas. Optimizing the pre-purification process, particularly in amine absorption, is critical to improving hydrogen production efficiency. Enhancements could focus on reducing energy consumption during the removal of hydrogen sulfide and carbon dioxide from natural gas, thus minimizing the energy load on downstream units, such as the reformer and PSA. In CLR, for example, streamlining the removal of contaminants early in the process can lower the energy demands of subsequent purification stages and reduce oxygen carrier degradation, improving overall system performance.

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